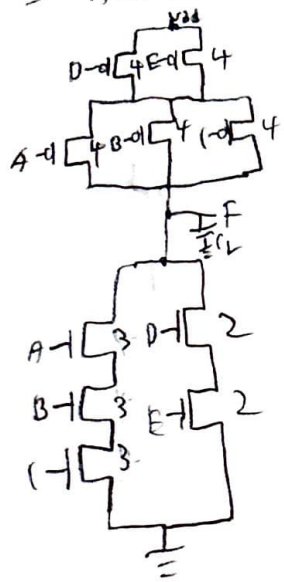
ECEN 153 HW 5 Dylan Thornhill

1. $F = \overline{ABC + DE} = \overline{ABC} \cdot \overline{DE} = (\bar{A}+\bar{B}+\bar{C})(\bar{D}+\bar{E})$

a+b.



$\frac{R_n}{R_p} = 2$

c. best case $t_{pHL} = .69(\frac{1}{2}R_n)C_L$

$t_{pLH} = .69(\frac{5}{24}R_p)C_L$

$(\frac{1}{3}+\frac{1}{3}+\frac{1}{3}) \parallel (\frac{1}{3}+\frac{1}{3}) = \frac{1}{2}$

$(\frac{1}{4} \parallel \frac{1}{4}) + (\frac{1}{4} \parallel \frac{1}{4} \parallel \frac{1}{4})$

$\frac{1}{8} + \frac{1}{12} = \frac{5}{24}$

d. $A=0,\ B=1,\ C=0,\ D=1,\ E=1$

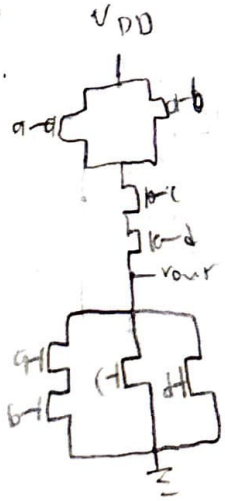
i. $F = \overline{0(1)0 + (1)(1)} = \overline{1} = 0$

ii. $R = \infty\Omega$

iii. $R = \frac{R_n}{2} + \frac{R_n}{2} = R_n$

2.

a.



b. $F = \overline{ab + c + d}$

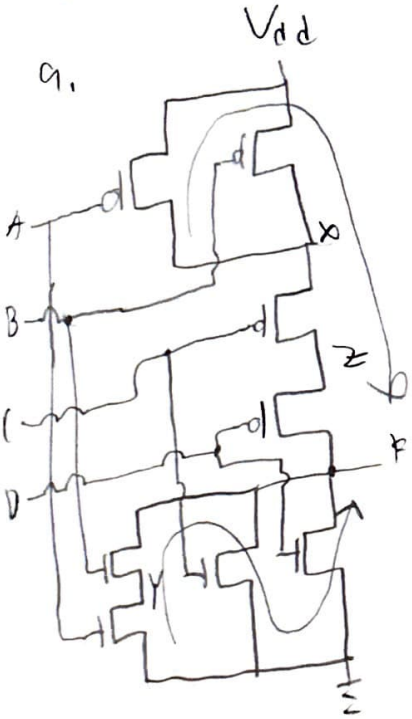
c. CMOS logic family

d. Full pullup and full pull down

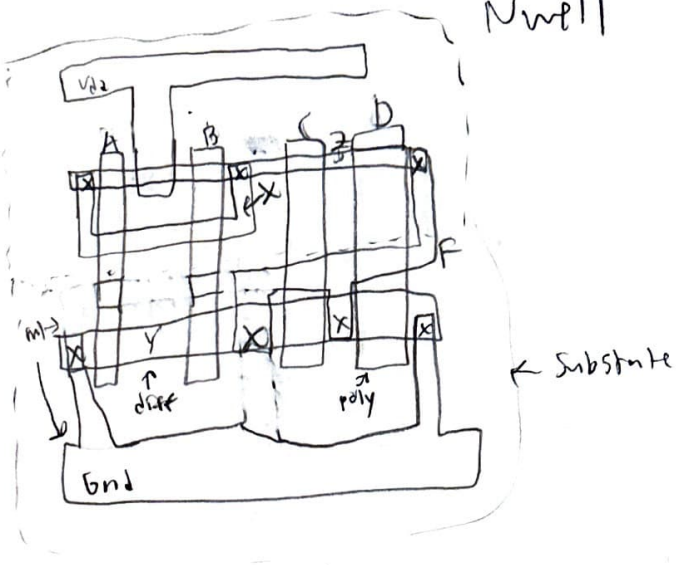
$V_{dd} - 0 = \boxed{V_{dd}}$

3.

a.


Vdd
A
B
C
D
X
Z
F
Y

b.


Nwell
Vdd
A
B
C
Z
D
X
F
Y
X
X
X
X
X
m1
diff
poly
Substrate
Gnd